SIMATS ENGINEERING

SIMATS Saveetha Institute of Medical And Technical Sciences (Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-9

Experiment 9: Travelling Salesman Problem (TSP)

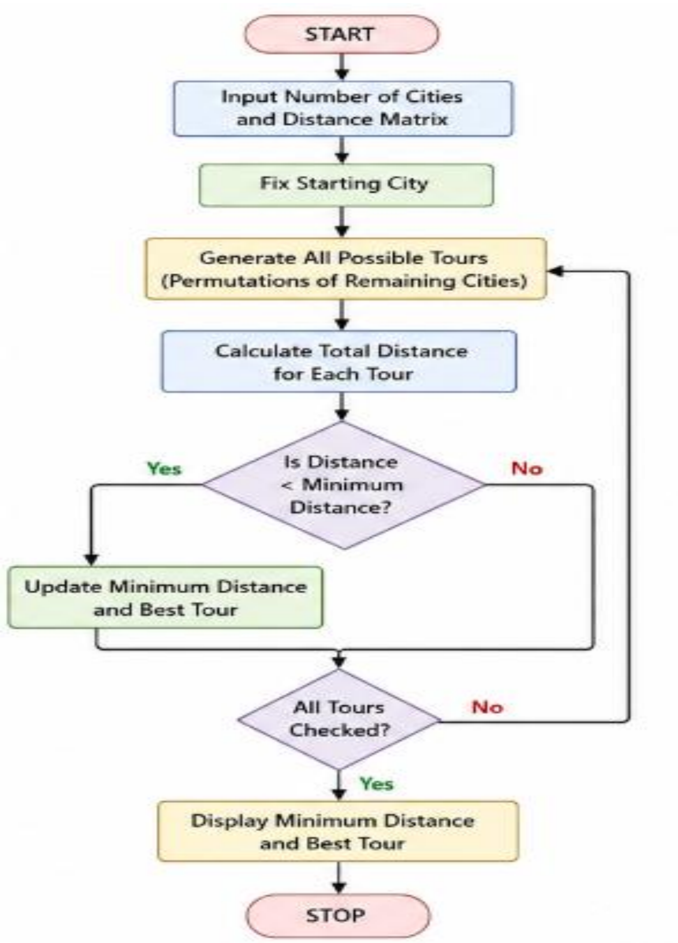
Aim

To implement the Travelling Salesman Problem using Python.

Objective

To find the minimum-cost path that visits every city exactly once and returns to the starting city.

Flowchart



Algorithm

1. Define the distance matrix.
2. Generate all possible routes.
3. Calculate the total distance for each route.
4. Compare all route costs.
5. Display the minimum-cost route.

Code

```
from itertools import permutations

graph = [
    [0,10,15,20],
    [10,0,35,25],
    [15,35,0,30],
    [20,25,30,0]
]

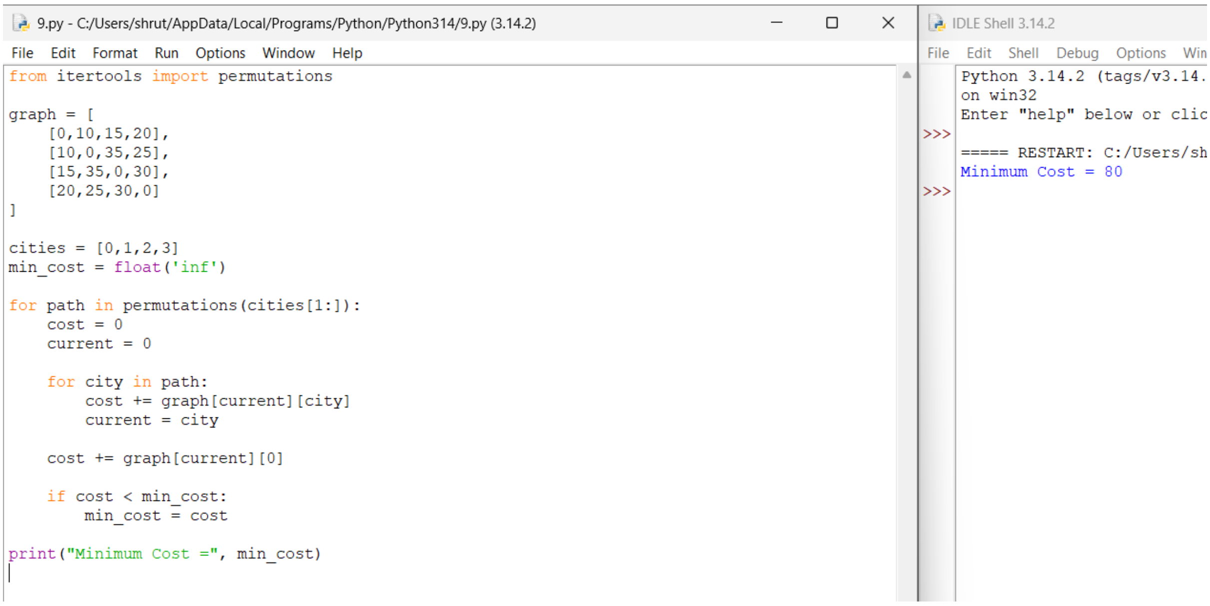
cities = [0,1,2,3]

min_cost = float('inf')

for path in permutations(cities[1:]):
    cost = 0
    current = 0
    for city in path:
        cost += graph[current][city]
        current = city
    cost += graph[current][0]
    if cost < min_cost:
        min_cost = cost

print("Minimum Cost =", min_cost)
```

Output



```
9.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/9.py (3.14.2)
File Edit Format Run Options Window Help
from itertools import permutations

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```
IDLE Shell 3.14.2
File Edit Shell Debug Options Win
Python 3.14.2 (tags/v3.14.
on win32
Enter "help" below or clic
>>>
===== RESTART: C:/Users/sh
Minimum Cost = 80
>>>
```

Result

The shortest route was successfully determined.

Conclusion

The Travelling Salesman Problem was solved by evaluating all possible routes and selecting the minimum-cost tour.